

Global analysis of next-generation utility-scale PV: Tracking bifacial solar farms

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HIGHLIGHTS

- Yearly energy gain of EW tracking over NS fixed-tilt 25–45% for latitude below 30°.
- EW tracking vs. NS tracking: 5–15% energy gain below 50° N/S and –5% above 50° N/S.
- EW tracking better than NS tracking for summer months and vice versa for winter.
- In-depth analysis on practicality of Sun-tracking vs. Power-tracking algorithms.
- Global analysis for minimum LCOE shows a worldwide optimum pitch of 2–3 m.

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ABSTRACT

The bifacial gain of East-West vertical and South-facing optimally-tilted bifacial solar farms are well established. One wonders if bifacial gain and the associated levelized cost of energy (LCOE) may be further improved by tracking the sun. Tracking bifacial photovoltaics (PV) system has advantages of improved temperature sensitivity, enhanced diffuse and albedo light collection, flattened energy-output, and reduced soiling. Monofacial tracking already provides many of these advantages, therefore the relative merits of bifacial tracking are not obvious. In this paper, we use a detailed illumination and temperature-dependent bifacial solar farm model to show that bifacial tracking PV delivers up to 45% energy gain when compared to fixed-tilt bifacial PV near the equator, and ~10% bifacial energy gain over tracking monofacial farm with an albedo of 0.5. An optimum pitch further improves the gain of a tracking bifacial farm. Our results will broaden the scope and understanding of bifacial technology by demonstrating global trends in energy gain for worldwide deployment.

1. Introduction

The global energy demand is approaching unprecedented levels with the projected population of ~10 billion people by 2050 [1]. Protecting the environment while meeting the energy demands will involve a delicate balancing act in the coming decades. It has been suggested that renewable energy resources can satisfy the energy demand with the smallest environmental footprint. Among all renewable energy resources, the direct conversion of solar energy to electricity by photovoltaic technology (PV) is one of the most promising sources of green

and sustainable energy with the advantage of limited environmental impact and low maintenance cost.

One way to reduce the levelized cost of energy (LCOE) is to design a photovoltaic (PV) system to produce the maximum energy for a specified area of the solar farm. Among the cost-effective options, one can either increase the total irradiance of fixed-tilt farms using bifacial modules or increase the normal irradiance on monofacial modules by solar-tracking. Both technologies are becoming popular as bifacial modules are expected to have ~40% market share by 2028 [2], whereas solar-tracking has been implemented by ~70% of the newly installed

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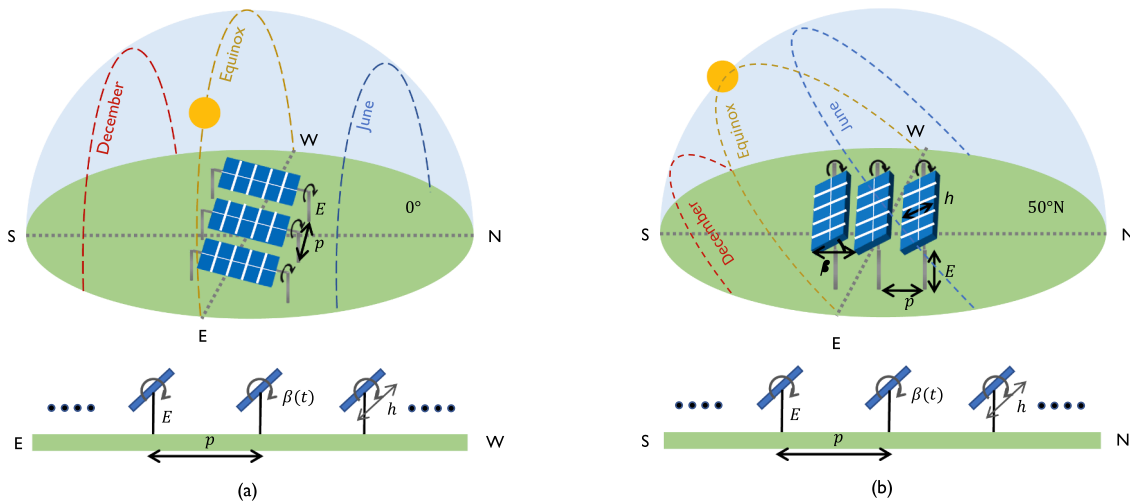


Fig. 1. Schematic of a single-axis tracking bifacial solar PV farm (a) E/W facing (b) N/S facing. The solar path varies with latitude as well as time making one of the designs geographically more suitable.

utility-scale PV systems since 2015 [3,4]. Indeed, market analysis from multiple groups worldwide suggest that single-axis tracking may account for 25–40% of all new utility-scale PV installation, motivated in part by the adoption of bifacial modules [5].

A key concern, however, is that the location-specific viability of bifacial tracking has not been quantified. The solar-tracking of bifacial PV modules sounds appealing, but the only concern is that whether bifaciality would be beneficial in solar-tracking configuration, as maximizing direct beam will cast a deep shadow on the ground and reduce the contribution from the ground albedo. Fortunately, the recent location-specific case-studies predict the promising potential of bifacial PV farms in solar-tracking configuration [6,7]. For example, the detailed calculation by Egido et al. [8] predicts that a *standalone dual-axis tracking* bifacial PV module over a white-painted ground has almost 80% higher annual energy yield as compared to its fixed monofacial counterpart. For the same white-paint albedo, a 20–90% increase in energy yield was demonstrated across different months for *single-axis tracking* of standalone bifacial PV modules as compared to fixed-tilt bifacial and monofacial modules, respectively. Similarly, Oria and Sala reported in [9] a 70% increase in annual energy yield using single-axis tracking of standalone bifacial modules with highly reflective albedo collectors as compared to their fixed monofacial counterparts. Shoukry et al. [10] demonstrated that both bifaciality and tracking have their own unique contributions depending upon the geographical locations. Compared to a fixed-tilt monofacial farm, bifacial tracking has a 44% gain and monofacial tracking has an 18% gain. There are other studies in the literature that demonstrate the promising potential of standalone bifacial and/or tracking PV panels both experimentally [11,12] and computationally [6,13]. There is an emerging consensus that the irradiance incident on a module can be enhanced through by combining bifaciality and tracking.

Despite these promising initial results for standalone modules, one still wonders if temperature-dependent efficiency degradation due to additional solar irradiance would erase any performance gain. Indeed, several research groups have explored the temperature-dependent performance of stand-alone PV modules [14,54–56]. By combining the irradiance, light collection, and electro-thermal models, several research groups have demonstrated that the bifacial gain and corresponding reduction in LCOE depend on the geographical location and one must optimize the module tilt and row-separation to maximize the bifacial gain [14,15].

Along with standalone PV modules, several research groups investigated utility-scale monofacial/bifacial PV farms both in fixed and/or sun-tracking configurations. Stefano et al. [16] demonstrated that the

bifaciality of single-axis sun-tracking PV arrays could enhance their performance by 12%. Stein et al. [17] reported ~10% bifacial gain for single-axis mobile PV systems. By using their mathematical model, Janssen et al. [57] demonstrated that single-axis tracking of monofacial and bifacial PV arrays could increase energy yield up to 15% and 26% respectively as compared to standard monofacial fixed-tilt system. A number of studies also reported the temperature-dependent performance of tracking PV farms based on theoretical analysis or simulated results [18–20]. Recently, based on their optical-electrical-thermal model, Patel et al. [21] showed that temperature-dependent efficiency could change LCOE of fixed-tilt bifacial utility-scale solar farms by –10% to 15% worldwide for different module technologies. Using a collection of solar farms in Brazil, Verissimo et al. [22] suggest that country-specific tracking results show solar-farm topology can have a significant impact on the energy yield and LCOE of the system. Rodriguez-Gallegos et al. [23] have also shown that the single-axis tracking bifacial PV system exhibit minimum LCOE worldwide, even though dual-axis tracking bifacial PV systems produce higher energy. A dual-axis system, however, has higher LCOE because its higher energy yield is partially counterbalanced by its higher initial, operational, and maintenance costs as compared to that for the single-axis system. After all, the single-axis system is simpler to design and operate both in E/W facing orientation to track the sun daily and in N/S facing orientation to track the sun over the seasons. Without taking into account location-specific land cost, the study [23] showed that single-axis bifacial PV farms exhibit a 16% decrease in LCOE for locations at latitudes within a range of $\pm 60^\circ$. These results further demand a careful evaluation of global-scale LCOE for single-axis tracking bifacial PV farms by considering the site-specific land costs. Indeed, as discussed in Ref. [24], the land-cost dramatically changes the optimum spacing, and the energy yield of a solar farm.

This work is the culmination of our series of studies on *standalone* and *farm-level* bifacial PV design considerations. We performed worldwide analyses investigating the optimum design parameters for energy maximization of a single bifacial PV module [14], a vertical bifacial PV farm [25], a ground-sculpted bifacial PV farm [26], and cost minimization of a fixed-tilt bifacial PV farm [21,24]. The purpose of this paper is to investigate the worldwide performance of utility-scale bifacial single-axis tracking PV farms (see Fig. 1) with fixed-tilt bifacial PV farms. We compare the performance comparison of two different tracking algorithms termed as *tracking the sun* (TS) and *tracking the best orientation* (TBO) [27]. By using a self-consistent optical-electrical-thermal model, we address the following key questions: (i) How do the energy yields compare between east/west (E/W) facing and north/

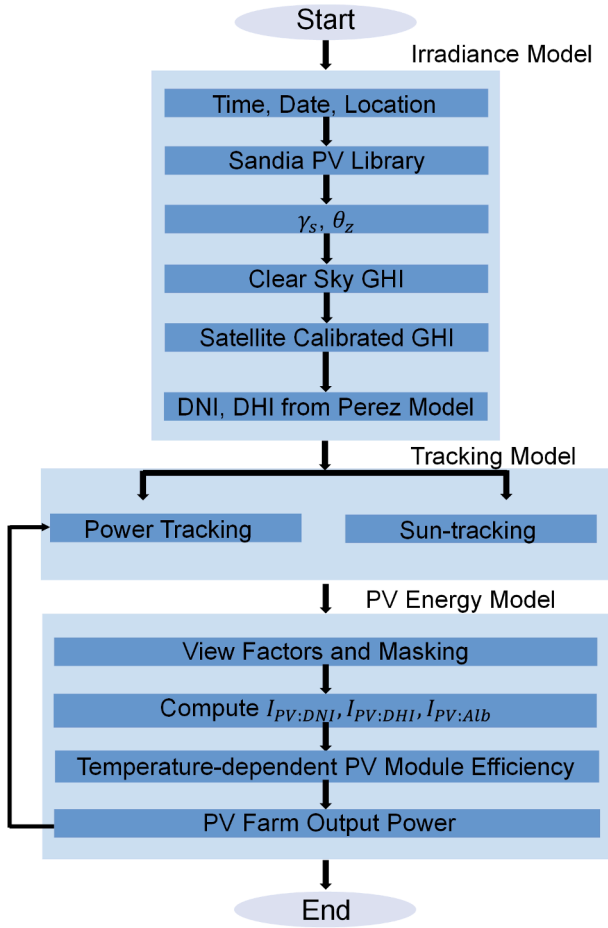


Fig. 2. The simulation workflow integrates irradiance, light collection, and tracking models to calculate the energy yield of a bifacial tracking solar farm.

south (N/S) facing bifacial single-axis tracking solar farms? (ii) How does the site-specific climate affect the performance of these farms? (iii) How do the location-dependent land costs affect the optimum design and LCOE of these farms? and (iv) What is the optimum row-spacing (pitch or period) for minimum LCOE?

This paper is divided into four sections. Section 2 describes the modeling approach. Section 3 discusses the results and the conclusions are provided in Section 4.

2. Modeling framework

To model a tracking solar PV farm, one must generalize and integrate several sub-models developed in previous studies [14,21,24–26]. We begin by integrating the irradiance model and a tracking algorithm to find the intensity of direct light falling on the solar modules. Next, we used a light collection model with view factors and masking formulae to estimate the diffuse and albedo light collection. Finally, the thermal model combined with the PV energy (electrical) model produces the total energy generated by the solar PV farm. A flow diagram in Fig. 2 demonstrates the complexity of a self-consistent model necessary to predict the performance of a tracking solar farm.

2.1. PV farm power and energy model

As shown in Fig. 1, we model a solar farm with the following design parameters: azimuth angle (γ_A) from the North; panel height (h); panel angle (β); elevation from the ground (E); pitch (p); and ground albedo (R_A).

Table 1

Equations associated with the Opto-Electrical Model: irradiance modeling and light collection.

(1) $I_{GHI} = I_{DNI} \times \cos(\theta_z) + I_{DHI}$	(6) $I_{PV:Alb,diff}^F(l) = I_{Gnd:DHI} \times R_A \times F_{dl-gnd}(E, l) \times \eta_{diff}$
(2) $I_{PV:DNI}^F(l) = I_{DNI} \cos(\theta_F(1 - R(\theta_F)) \eta_{dir}; l \text{ is unshaded}$ $= 0; l \text{ is shaded}$	(7) $I_{PV:Alb,diff}^F = 1/h \int_0^h I_{PV:Alb,diff}^F(E, l) dl$
(3) $I_{PV:DHI}^F = I_{DHI} \eta_{diff} / h \int_0^h \frac{1}{2} (1 + \cos(\psi(E, l) + \beta)) dl$	(8) $I_{PV:Alb}^F = I_{PV:Alb,dir}^F + I_{PV:Alb,diff}^F$
(4) $I_{PV:Alb,dir}^F(l) = I_{Gnd:DNI} \times R_A \times F_{dl-gnd}(E, l) \times \eta_{diff}$	(9) $I_{PV(T)}^{Total} = I_{PV:DNI}^F + I_{PV:DHI}^F + I_{PV:Alb}^F$
(5) $I_{PV:Alb,dir}^F(l) = 1/h \int_0^h I_{PV:Alb,dir}^F(E, l) dl$	(10) $YY_T(p, \beta, h, E, \gamma_A, R_A) = \int_0^1 I_{PV(T)}^{Total}(p, \beta, h, E, \gamma_A, R_A) dY$

Table 2

Equations associated with the Thermal Model.

(11) $\eta(T_{cell}) = \eta_{STC}(1 - TC(T_{cell} - T_{STC}))$
(12) $T_{cell} = T_{amb} + C_T \frac{\tau_{POA}}{U_L} \left(1 - \frac{\eta(T_{cell})}{\tau_{POA}} \right)$
(13) $I_{POA} = I_{PV:DNI}^F / \eta_{dir} + (I_{PV:DHI}^F + I_{PV:Alb}^F) / \eta_{diff}$

First, we use the **irradiance model** and estimate the ground illumination for a particular time of the day with a time step of 1 min at a location specified by its latitude and longitude. We follow the modeling approach described in [21,24,25]: We use the PV library from Sandia National Laboratory to calculate the Sun's trajectory (zenith (θ_z) and azimuth angle (A)) and the irradiance [25]. The Global Horizontal Irradiance (GHI or I_{GHI}) is ideally given by Haurwitz clear sky model [28,29]. Renormalizing this irradiance based on the NASA Surface meteorology and Solar Energy database [30] yields the local variation in GHI. The 22-year monthly average GHI data from NASA Surface meteorology and Solar Energy database has a spatial resolution of $1^\circ \times 1^\circ$ (latitude $\times$ longitude). The GHI is then split into direct light (DNI or I_{DNI}) and diffuse light (DHI or I_{DHI}) using Orgill and Hollands model [31]. Perez model [32] further improves the accuracy by estimating both the circumsolar and isotropic components of diffuse light. The bifacial module output based on this insolation models has been shown to closely match experiments [14].

Second, we quantify the amount of **light collected by the solar panels** installed at that location. The panels have height h , tilted at an angle β , separated by pitch (or period) p , and are oriented at an array azimuth angle $\gamma_A = 180^\circ$ (i.e., south-facing panels) for farms in the northern hemisphere and $\gamma_A = 0^\circ$ (i.e., north-facing panels) for farms in the southern hemisphere or $\gamma_A = 90^\circ$ for panels facing E-W direction. The collection of light on panels from the three components of irradiance (i.e., direct, diffuse, and albedo) are formulated separately and analyzed accordingly. Our approach to model the collection of light involves the view-factor calculation described in [21,24]. The corresponding equations are summarized in Table 1 with a glossary of the terms defined in Table 3. A correctly implemented view-factor model gives identical results compared to the ray-tracing approach [58].

Third, we incorporate the **temperature-dependent efficiency model** which require the collected light intensity, efficiency at STC (Standard Testing Conditions), and ambient temperature as inputs and yield the effective efficiency as output [21]. The temperature-dependent efficiency loss involves a complex interplay of increased absorption due to bandgap reduction vs. increased dark-current and reduced mobility. At practical illumination intensity, it is well-known that the efficiency $\eta(T)$ scales linearly with the module/cell temperature, see Eq. 11 in Table 2 where the rate/slope of degradation is given by the absolute

Table 3
Glossary of the symbols used in this paper.

Parameters	Definition
TC	Temperature Coefficient (%/°C)
Tamb	Ambient Temperature (°C)
Tcell	Cell Temperature (°C)
η	Power conversion efficiency (%)
I/Irradiance	Input optical power over the plane of array (W/m ²)
c _T	Correction term used for the daily average temperature data
τ	Transmittance of glazing (no unit)
α	Absorbed fraction (no unit)
u _L	Heat loss coefficient (W/m ² K)
θ_z	Zenith Angle (<i>degrees</i>)
θ_f	Angle of incidence at the front face of the panel
Pitch (p)	Row-to-row distance between the bottom edges of consecutive arrays (m)
Height (h)	Height of the panel
E	Elevation
β	Tilt angle
γ_A	Azimuth angle from measure from the North
R _A	Albedo
C/C	Cost / Cost per unit meter
M	Number of rows/arrays of modules
Z	Number of modules in an array
Y	Lifetime of a farm (in years)
YY	Yearly Yield
Sub/Super-scripts	Definition
amb	Ambient
Alb	Albedo
DHI	Diffuse Horizontal Irradiance
diff	Diffuse
dir	Direct
dl	Differential element along the height of a panel
DNI	Direct Normal Irradiance
F	Front
f	Fixed cost
Farm	PV Farm
GHI	Global Horizontal Irradiance
Gnd	Ground
M, cell	Module/cell
Panel	PV panel
PV	Photovoltaic
STC	Standard testing condition
Total	Total
Z	Zenith

temperature coefficient (TC) [33,34].

Finally, we find **the daily, monthly, and yearly energy-output** of the farm. Using the equations (2), (3), and (8) in Table 1, we arrive at Eq. (9) to find the time-varying spatially distributed light collection on the panels. This information combined with the thermal model (summarized in Table 2) is used in the circuit model to find the equivalent power generation. The circuit model includes the bypass diode model to take into account the partial shading of the PV panels and their effect on the PV performance [25,35]. To estimate energy output, we integrate the power generated over the desired period of time. We define the energy yield per pitch of a farm over one year as yearly yield (YY) given by Eq. (10). The associated levelized cost of energy (LCOE) is modeled and estimated using the concept of essential levelized cost of energy (LCOE*) as detailed in Refs. [24,36]. The following equation describes the relation between LCOE and LCOE* where M_L is the ratio of module-related C_M and land-related costs C_L . χ is a constant that includes the degradation rate and bank discount rate.

$$LCOE^* \equiv \frac{LCOE \cdot \chi}{C_L} = \frac{C_M/C_L + p/h}{YY(p, h, E, \beta, \gamma, R_A)} = \frac{p/h + M_L}{YY} \quad (14)$$

2.2. Tracking model

In a PV tracking system, instead of a fixed panel tilt, the tilt angle $\beta(t)$ varies with time. There are two types of tracking algorithms: (a) **sun-**

tracking orients the modules to maximize the direct light, and (b) **power-tracking** orients the modules to maximize power or energy.

As the name suggests, a **sun-tracking** system minimizes the angle between the sun and the panel. To compute the time-dependent tilt angle of E/W facing modules by the **sun-tracking algorithm**, first, we compute the angle of incidence (θ_{AOI}) between the direct light (DNI) and the normal of the module face. Solar tracking (TS) minimizes the angle of incidence towards 0° so that the face of the solar panel follows the sun (direct light). Fig. 3 shows the hourly variation of the tilt angle β for TS for June and December at Washington D.C. (38.9°N, 77.0°W) and Dubai (24.5°N, 55.5°E) in the northern hemisphere.

For E/W tracking, the panels follow the sun throughout the day, and thus the tilt angle $\beta(t)$ goes from +90° (East-facing) to -90° (West-facing) from sunrise to sunset. For N/S tracking, during summer (June) around sunrise, $\beta(t)$ is negative i.e., North-facing panels with $\gamma_A = 0^\circ$ becomes close to 0° at solar noon and goes back to negative during sunset. This is because the sun's path, from sunrise to sunset, lies almost entirely in the North except near solar noon where the sun could be in the south of the panels. Whereas, during winter (December), $\beta(t)$ remains always positive i.e., South-facing panels with $\gamma_A = 180^\circ$ because the sun's path is entirely in the South. Further, the panels face the south-inclined sun normally at solar noon where the $\beta(t)$ is equal to the zenith angle of the sun. Thus, the sign of tilt angle (and the array azimuth angle) varies with the sun's path according to the latitude of a

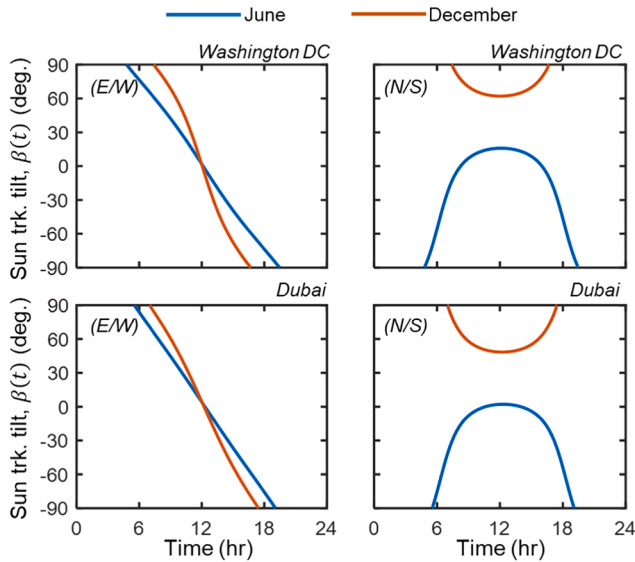


Fig. 3. Using the sun-tracking algorithm, we plot the tilt angle as a function of time for E/W facing tracking PV modules (left column) and N/S facing tracking PV modules (right column) for June (Blue) and December (Red) at Washington, D.C., 38.9°N, 77.0°W (top row) and Dubai, 24.5°N, 55.5°E (bottom row). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

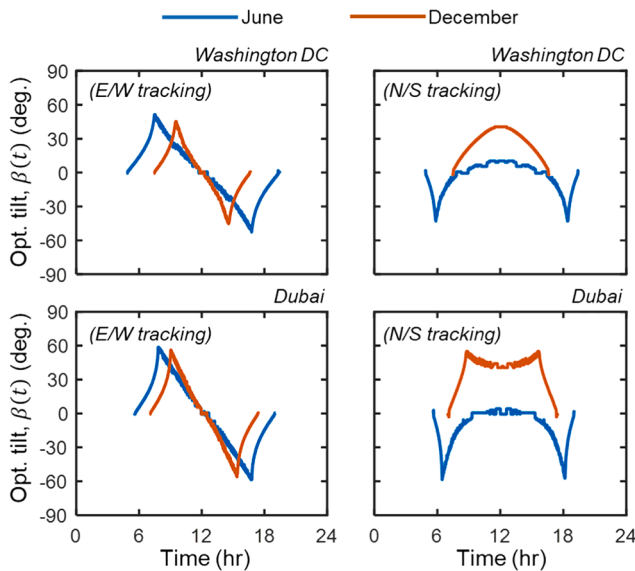


Fig. 4. Using the power-tracking algorithm, we plot the tilt angle as a function of time for E/W facing tracking PV modules (left column) and N/S facing tracking PV modules (right column) for June (Blue) and December (Red) at Washington, D.C. (top row) and Dubai (bottom row). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

geographical location throughout the year. The days in December are shorter than the days in June, which is evident from the sunrise and sunset points. Moreover, $\theta_{AOI} = 0^\circ$ is not achievable for both months, thus, the algorithm minimizes θ_{AOI} for the closest normal incidence. Finally, the initial angle being $\pm 90^\circ$ is impractical for a solar farm because in this scenario the first row of panels will cast a row-to-row shadow on all the other panels.

The second approach involving **power-tracking** optimizes the time-dependent tilt angles to achieve maximum power at each time-step—this will also maximize daily energy. The optimization is governed by

the combined effects of (a) relative contributions of instantaneous DHI and DNI, (b) position of the sun, and (c) panel-to-panel shading. This power-tracking or the best orientation (TBO) algorithm is targeted towards maximizing output, not the light collection. Thus, during the early and late part of the day, the system will compromise light collection to avoid shading loss and maximize power output.

Fig. 4 shows the time-dependent optimized tracking angles $\beta(t)$ in June and December for Washington D.C. and Dubai. During the early and late parts of the day, the irradiation is mostly sky-diffuse light. At these times, in contrast to the TS algorithm, the panel optimally faces the sky (i.e., $\beta(t) \approx 0$) for both E/W and N/S tracking configurations. As the day progresses, the fractional contribution of DNI increases, and $\beta(t)$ will compete between facing the sky and the sun. This is prominently observed for optimal tilt for E/W tracking panels in Fig. 4. After a certain time, e.g., 7 am in June, the panels closely follow the sun. In general, even when the DNI fraction is higher, the $\beta(t)$ is shallower than sun-tracking angles to improve the collection of DHI. The N-shape profile of $\beta(t)$ in E/W tracking is narrower in winter seasons due to shorter days. For N/S tracking, $\beta(t)$ is larger in December compared to June due to the more tilted sun-path.

To compare the tracking algorithms in terms of insolation collection, Rodriguez-Gallegos et al. [27] performed a PV tracker performance analysis using these two algorithms – tracking the sun (TS) and the best orientation (TBO) – for 3 different configurations, namely, horizontal single-axis tracker (HSAT), tilted single-axis tracker (TSAT), and dual-axis tracker (2 T). They found that TBO shows a higher insolation collection compared to TS but the difference is below 1.8% for $|\text{latitude}| < 60^\circ$ and 3.3%, 7.1%, and 2.9% for HSAT, TSAT, and 2 T, respectively.

A few points regarding practical tracking algorithms are noteworthy. First, the power-tracking or TBO algorithm requires the panels to backtrack at least twice during the day (observe the peaks and troughs in Fig. 4). This may cause additional mechanical wear and tear of moving parts and electrical issues with the motor. In contrast, the sun-tracking algorithm continuously follows the sun during the day which avoids backtracking and related reliability issues, if any. Second, the maximum value of the tilt angle for tracking the panels is 60° to avoid rotational load on the axis and avoid mutual shading between the rows of panels at higher tilt angles. Third, in practice, the panels are kept at an initial tilt of 60° (not horizontal at 0°) to reduce dew-enhanced soiling that would reduce the output of the solar panels.

In the following section, we will discuss the energy yield of solar farms based on the power-tracking algorithm.

3. Power and energy yields at specific locations

In this section, we will summarize the modeling and simulation results through energy yield and then discuss LCOE-optimized solar farms. For an intuitive understanding of the results, we first compare the monthly energy output for various farm designs at four locations – Dubai (24.5°N, 55.5°E), Washington, D.C. (38.9°N, -77.0°W), Stockholm (59.3°N, 17.8°E), and Sydney (33.8°S, 150.6°E). Next, we present the global energy yield maps for fixed-tilt, East-West (E/W), and North-South (N/S) tracking farm designs. By default, we assume an east-west (E/W) tracking bifacial farm with a pitch over height, $p/h = 2$ and elevation above the ground, $E = 1$ m (we will discuss pitch and height-optimized design later in Section 4). We then compare this default design with various other PV farm configurations, namely, fixed-tilt bifacial farm, E/W tracking monofacial farm, and N/S tracking bifacial farm. Finally, we will present the effect of varying pitch on the yearly energy yield and LCOE* for the default EW tracking PV farm design.

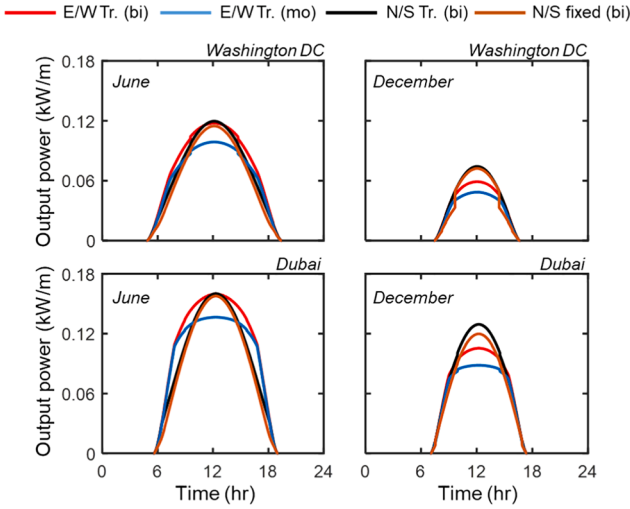


Fig. 5. Daily output power for Washington D.C. and Dubai during summer months (June) and winter months (December). E/W tracking performs better in summer months and lower latitudes while N/S tracking has a higher output during winter months and higher latitudes.

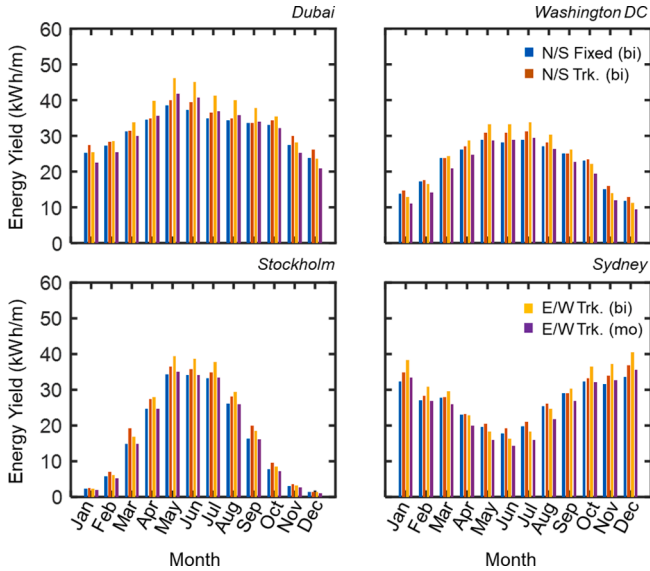


Fig. 6. Monthly energy yield for various farm designs, namely, N/S Bifacial Fixed tilt (Blue), N/S Tracking (Red), E/W Bifacial Tracking (Yellow), and E/W Monofacial Tracking (Purple). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.1. Power output

E/W tracking vs. N/S Fixed-tilt Systems. The power outputs of the PV farm using the power-tracking algorithm along with N/S fixed-tilt systems are shown in Fig. 5. We find that the energy yields of E/W tracking bifacial farms are always higher than that of E/W tracking monofacial farms for Dubai and Washington, D.C. In June (for both Washington DC and Dubai), E/W tracking yields higher energy compared to N/S tracking systems slightly before- and after-noon, resulting in overall higher output from E/W tracking PV. In December, as the sun-path is more tilted, the E/W systems cannot adequately follow the sun (θ_{AOI} is high even at noon). Thus, the N/S tracking and fixed-tilt systems give more output than E/W tracking systems in December. Although the net gain depends on the location, the counterbalancing energy yield for summer and winter holds for any location, as discussed in the next

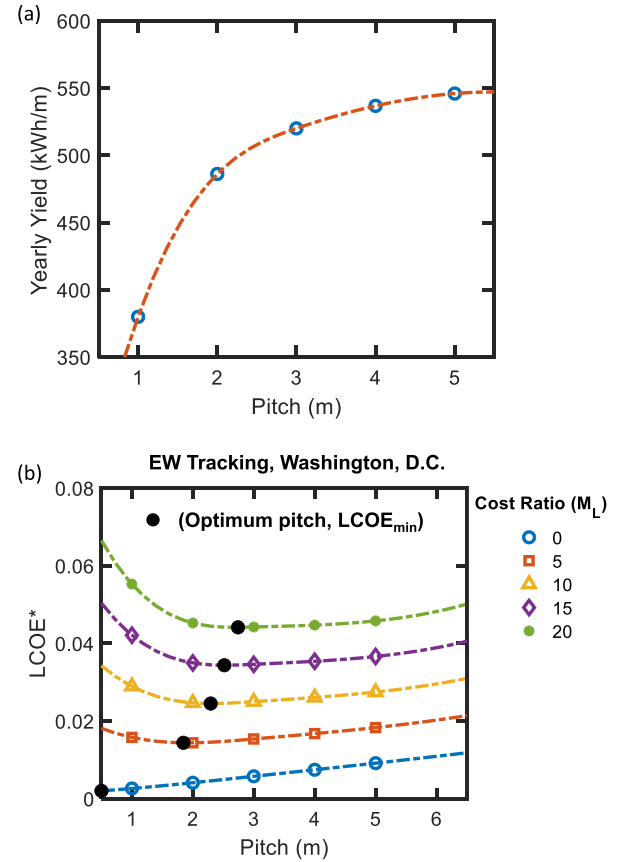


Fig. 7. (a) An increasing trend in yearly energy yield with increasing pitch. The yearly energy yield saturates at higher values of pitch. (b) For a given value of cost ratio (M_L), an optimum pitch provides a minimum LCOE*. The optimum pitch increases with increase in module to land cost ratio (M_L).

section.

N/S tracking vs. N/S Fixed-tilt Systems. The N/S tracking panels have a higher output than N/S fixed-tilt panels as it can avoid shading in mornings and evenings and correct for panel tilt for seasonal sun-path variations. For example, in Fig. 5(bottom-right) there is a clear effect of partial shading in Washington DC in December for N/S fixed panels (observe the lowered output before 9 am or after 3 pm).

3.2. Monthly/Seasonal energy yield variation

The monthly energy yield for various PV module configurations is shown in Fig. 6. Various locations in both the northern and the southern hemispheres are explored here. During summer months in the respective hemispheres, E/W bifacial solar tracking scheme performs best, whereas, in the other months, N/S bifacial solar tracking scheme performs better. One of the important points to note is that N/S bifacial fixed-tilt performs best during Nov. to Jan. in Stockholm. Here, although higher irradiance is collected by bifacial modules for N/S solar tracking configuration, higher module temperature lowers its efficiency as compared to its fixed-tilt counterpart. Moreover, the average monthly energy gains for Dubai, Washington, D.C., Stockholm, and Sydney, respectively are as follows -

- (i) E/W tracking vs. N/S tracking: 5.75%, 0.16%, -3.00%, and 1.08%.
- (ii) E/W tracking bifacial vs E/W tracking monofacial: 11%, 15%, 18%, and 13%.
- (iii) N/S tracking vs. N/S fixed tilt: 6%, 5%, 14%, and 6%.

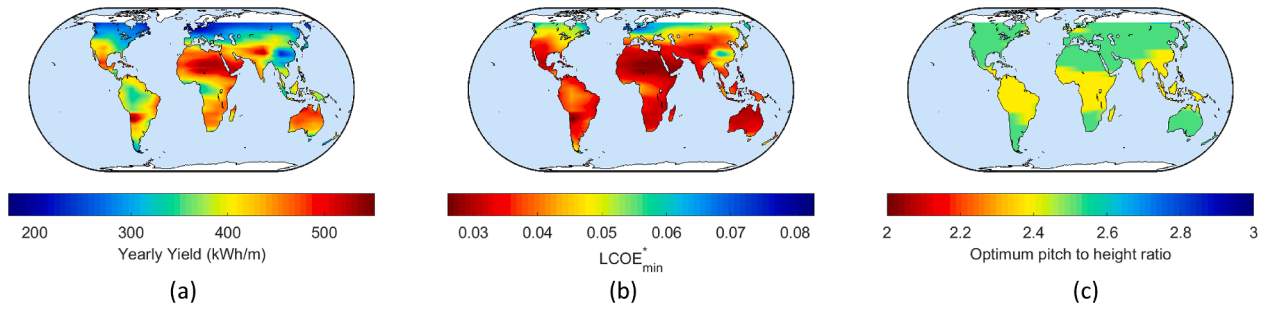


Fig. 8. Global trends in yearly energy yield, LCOE*, and optimum pitch/height of an elevated bifacial farm ($E = 1m$) for albedo $R_A = 0.5$ and $M_L = 12$.

The yearly minimum and maximum energy gains are shown in Fig. S4. The minimum is observed in the winter months and maximum in the summer months of the respective hemisphere. In terms of energy yield, it is apparent from these results that E/W tracking would be better near the equator and N/S tracking would be beneficial for higher latitudes.

We have so far used a fixed $p/h = 2$ for the daily and monthly energy calculations, however, the same p/h would not generate the maximum energy for all the farm designs due to the difference in row-to-row shading throughout the day. Thus, for energy maximization and cost minimization, we need to find the optimum pitch over height ratio (p/h) for specific farm design at a given location, as discussed in the next section.

3.3. Optimum p/h , YY, and LCOE-min

The pitch of a solar farm – defined as the spacing between the rows of the PV panels – is an important design parameter. A farm is optimized by first quantifying how yearly energy yield and LCOE depend on the pitch of a solar farm.

For the daily power output and monthly energy calculations in the previous sections, we assumed the pitch over height ratio $p/h = 2$, for all the locations. In practice, higher p/h improves the energy yield (per panel) due to an increase in albedo light collections and reduction in mutual shading between the rows of solar panels, see Fig. 7(a). However, once land-cost is brought into consideration (in other words, when the land is limited), then an optimum value of p/h yields the minimum LCOE. This optimum p/h is depicted in Fig. 7(b) for an E/W-tracking system at Washington, D.C. We find that the optimum p/h increases with increasing module-to-land cost-ratio (M_L). For a given module-related cost, a decrease in cost-ratio implies an increase in the land-related costs, leading to a reduction in the optimum pitch. Moreover, the optimum pitch and the associated minimum value of LCOE varies with the geographical location.

4. Global-scale merits of various farm designs

Given the understanding of power and energy-yield at the four cities across the world, now we are ready to compare the technologies across the globe. As discussed in Ref. [24], however, we must optimize the row-spacing depending on the LCOE-considerations.

In recent years, the M_L has ranged from 9 to 15 [24,36]. Therefore, the optimum p/h for minimum LCOE lies between 2 and 3 for most locations around the world, see Fig. 8(c). For $M_L = 12$, the global yearly energy yield and minimum LCOE* at optimum p/h are shown in Fig. 8. (This result led to the default choice of $p/h = 2$ in Section 3.) Nonetheless, it is preferred that prior to the PV farm deployment, we optimize the farm design through an analysis of LCOE (proportional to LCOE*) variation with the pitch for each farm site around the world.

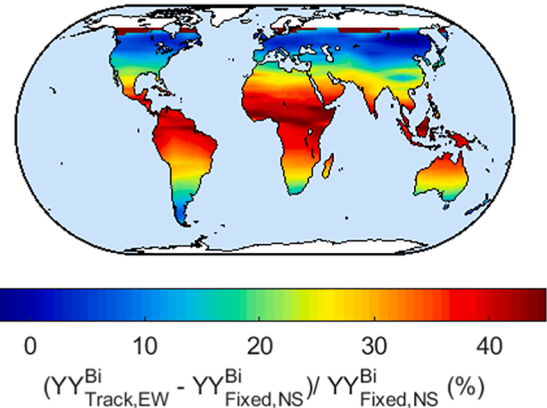


Fig. 9. Percentage gain in yearly energy yield in East-West tracking bifacial PV farm compared to North-South facing fixed-tilt bifacial farm. The energy gain for E/W tracking is highest near the equator and in the deserts around the world whereas fixed-tilt performs slightly better at very high latitudes. For this calculation, the albedo $R_A = 0.5$.

4.1. Energy yield of E-W tracking bifacial solar farms

Fig. 8(a) shows the yearly energy yield of an E/W tracking bifacial PV farm with an optimum pitch over height for $M_L = 12$. The yearly energy output per unit panel height reaches approximately 550 kWh/m for geographical locations close to the equator and deserts across the globe. These locations receive the highest fractions of direct light and thus, a tracking system benefits the most at these places. However, for locations with high diffuse light fraction, i.e., Beijing, Northern Europe, and North America, the yearly yield is significantly lower, 200 to 300 kWh/m. The E/W tracking PV farms are not advantageous in these regions. The yearly energy yield in Fig. 8 will be taken as a reference for the comparisons performed in the following sections.

4.2. Tracking PV farm vs. fixed-tilt bifacial farm

Let us now determine if a single-axis tracking PV farm should be a preferable choice of deployment as compared to a fixed-tilt farm at any specific location around the world. The energy surplus/deficit of a tracking PV farm over a fixed-tilt PV farm depends on the geographical location, see Fig. 9. The E/W tracking bifacial PV farm outperforms the N/S fixed-tilt bifacial PV farm by 45% near the equator and at lower latitudes. The highest energy gain is observed in desert regions, e.g., Sahara, Atacama, Australian, and Kalahari. These regions receive a high fraction of direct light irradiance and EW tracking PV relies on tracing the direct light, thus leading to increased energy gain. However, for sites with low direct light fraction and in turn high diffuse fractions, e.g., latitudes $> 45^\circ$ and Northeast China, the fixed-tilt bifacial farm is comparable or slightly better in energy gain. Given the additional land-area needed, higher initial and O&M cost, tracking may not be suitable

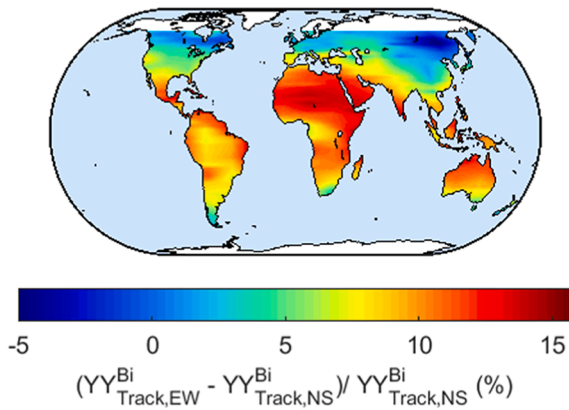


Fig. 10. Percentage gain in yearly energy yield in East-West tracking bifacial PV farm compared to North-South tracking bifacial PV farm. The energy gain for E/W tracking is highest near the equator and in the deserts around the world whereas N/S tracking performs better at very high latitudes. For this calculation, the albedo $R_A = 0.5$.

for these locations.

Moreover, a comparison between E/W tracking monofacial farm and N/S fixed-tilt bifacial farm shows that tracking monofacial farm is advantageous only for locations with high fractions of direct light i.e., $|\text{latitude}| < 30^\circ$, whereas N/S fixed-tilt bifacial farm performs better for higher latitudes and for regions where the diffuse light fraction is relatively higher. Fig. S1 clearly displays this trend.

Overall, it will come as no surprise that tracking PV farms are advantageous for locations with a high direct light fraction while fixed-tilt bifacial farms perform better at sites with the high diffuse light fraction. Our contribution is to quantify the energy gain associated with various configurations of optimized solar farms.

4.3. E/W vs. N/S tracking bifacial farm design

Next, we explore a conventional E/W single-axis tracking PV farm design with an N/S single-axis tracking one. In contrast to the hourly tracking of direct sunlight in an E/W tracking PV, an N/S tracking farm tracks the seasonal change in sun path (solar elevation/zenith angle) and varies comparatively less during the day. Since N/S fixed-tilt bifacial farm generates more energy output as compared to an E/W tracking farm at higher latitudes, we expect an N/S tracking farm to perform even better than a fixed-tilt bifacial farm.

Fig. 10 displays a worldwide comparison between an E/W tracking farm and an N/S tracking farm. As expected, we observe that N/S tracking farm to achieve higher energy output than E/W tracking farm for locations with high diffuse light fraction, e.g., Northern Canada and Siberia. Moreover, in comparison to E/W tracking farm, the (negative) percentage yearly energy gain is higher for an N/S tracking farm (Fig. 10) than an N/S fixed-tilt bifacial PV farm (Fig. 9) for higher latitudes.

Thus, with 15% energy gain, E/W single-axis tracking PV farm is a better design for locations with $|\text{latitude}| < 50^\circ$ and regions with a high direct light fraction. An N/S single-axis tracking farm on the other hand shows 5% higher energy yield than the E/W tracking design for sites with $|\text{latitude}| > 50^\circ$ and those with a high diffuse light fraction.

4.4. Does land-type matter for energy yield?

In the previous sections, we have analyzed the global energy yields for various tracking and fixed-tilt PV configurations. However, it is interesting to examine these solar farm designs in the context of the type of land of the installation site. This investigation may be especially useful for the new range of agro-photovoltaic (agrivoltaics) applications

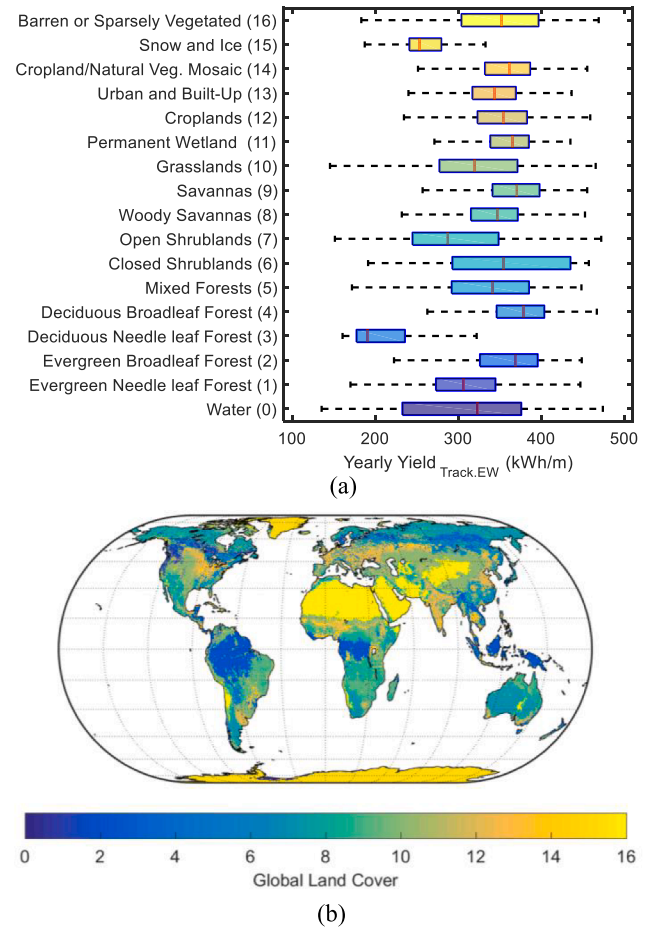


Fig. 11. (a) Yearly energy yield of E/W tracking bifacial PV farm over various types of land [41]. (b) A map of the 17 land types corresponding to the y-axis in (a). “0” corresponds to “Water” while “16” refers to “Barren or sparsely vegetated” land type. Albedo data is acquired from NASA MODIS [38–40].

around the world. For example, it has been recently claimed that croplands in moderate climatic conditions are preferred locations for solar farms as well [37].

In addition to the model described in Section 2, we further embed a monthly-average value of albedo for all the land types around the world. The global monthly albedo data is acquired from NASA MODIS [38–40] and yearly-average value of albedo for the 17 land types is tabulated in the appendix section S5. Fig. 11 summarizes our result of the analysis succinctly. We consider 17 types of land as shown on the y-axis of Fig. 11 (a) – the data was collected from Ref. [41]. These land types are also marked on the world map in Fig. 11(b), where “16” corresponds to “Barren or Sparsely Vegetated” land, and “0” refers to “Water”. We observe that, on an average, maximum energy yield is expected for the land types “2” (Evergreen Broadleaf Forest), “4” (Deciduous Broadleaf Forest), and “9” (Savannas). However, the land types (2 and 4) are rich in biodiversity and we should avoid installing PV farms here to preserve ecological balance. “Savannas” (9) and “Barren lands/Deserts” (16) show good potential for E/W tracking farm deployment in terms of yearly energy yield. Land type “3” (Deciduous Needle leaf Forest) and “15” (Snow and Ice) are the least favorite for tracking E/W bifacial PV. In summary, we can infer from Fig. 11 that installing E/W tracking bifacial PV farms in parts of Deserts, and Savannas are highly viable. The regions of croplands and grasslands have medium potential for energy yield (E/W tracking) and thus can be combined with agricultural usage and animal husbandry.

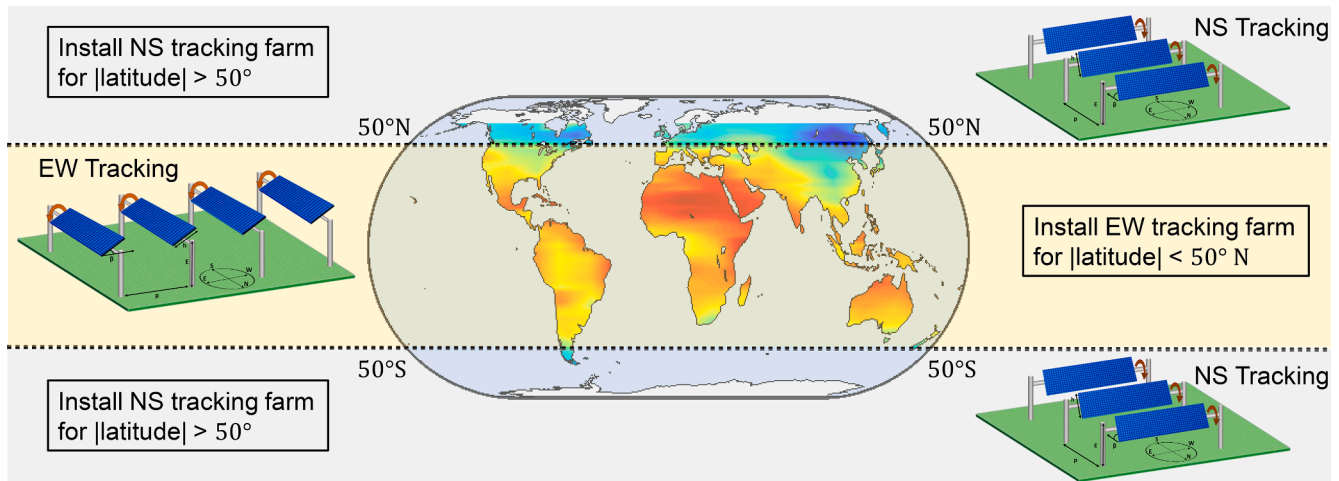


Fig. 12. In conclusion, EW tracking bifacial farms outperform NS tracking bifacial farms for $|\text{latitude}| < 50^\circ$ and vice-versa for $|\text{latitude}| > 50^\circ$.

5. Summary and conclusions

In this paper, we have investigated the energy yield and comparative performance of various solar farm design configurations such as single-axis tracking vs. fixed-tilt and bifacial vs. monofacial. Our modeling framework combines the irradiance model, light collection model, temperature-dependent efficiency model, and a single-axis tracking model to estimate the energy yield of a solar farm. The tracking algorithm constraints the panel tilt angle to vary such that the angle of incidence is 0° i.e., the direct light falls normally on the panel. These models are simulated for locations worldwide that finally output global maps which quantify the percentage change in energy yield while comparing various farm designs. The analysis of this paper leads to the following key conclusions:

- An E/W single-axis tracking bifacial PV farm has up to 45% higher yearly energy yield than an N/S facing fixed-tilt bifacial PV farm for locations at $|\text{latitude}| < 30^\circ$, as shown in Fig. 9. A higher fraction of direct light leads to higher energy gain. An E/W tracking monofacial farm vs. fixed-tilt bifacial farm shows similar global trends with maximum energy gain reaching $\sim 10\%$ near the equator (Fig. S1). However, fixed-tilt bifacial farm outperforms tracking monofacial farm by 5–15% for $|\text{latitude}| > 30^\circ$ (Fig. S2).
- As shown in Fig. 10, an E/W single-axis tracking bifacial PV farm tracks the hourly movement of the Sun whereas an N/S single-axis tracking bifacial farm mostly tracks the seasonal/monthly movement of the Sun. The former provides up to 15% more energy for locations close to the equator ($|\text{Latitude}| < 50^\circ$) and the latter generates up to 5% for locations close to the poles ($|\text{Latitude}| > 50^\circ$).
- Monthly/Seasonal variation in energy yield shows that the E/W tracking produces higher energy output in summer months whereas N/S tracking is favorable in winter months due to a southward inclination of the solar path in the Northern hemisphere and vice versa for the Southern hemisphere, as seen in Fig. 6.
- On an average, maximum PV energy yield is achieved at “Evergreen and Deciduous Broadleaf Forests” land types. However, to preserve the biodiversity at these forests, it would be ideal to install E/W bifacial tracking PV on the next best land type for high energy yield, namely, “Deserts” and “Savannas”, as illustrated in Fig. 11.
- The pitch over panel-height ratio (p/h) of an E/W tracking farm is optimized by minimizing LCOE. The value of optimum p/h increases with the module-to-land cost-ratio (M_L) and varies with the geographical location of the farm. Whereas the yearly energy of the farm, however, increases monotonically with p/h and saturates for very high values of pitch, as shown in Fig. 7. For a typical value of

$M_L = 12$, the optimum p/h lies between 2 and 3 globally, as seen in Fig. 8.

Fig. 12 summarizes our key conclusion that an E/W single-axis tracking bifacial PV farm is the best PV farm design for most regions ($|\text{Latitude}| < 50^\circ$) around the world. The energy gain with respect to a fixed-tilt bifacial or an N/S tracking farm varies according to the incident direct light fraction and the solar path at the geographical location of the solar PV farm. Moreover, for minimizing LCOE of a farm an optimum pitch can be used according to the estimated essential module to land cost ratio (M_L) for the deployment site. **Thus, in terms of energy maximization, a bifacial tracking PV would be a worthwhile farm technology in most locations of the world.**

In principle, the results are easily generalizable to any bifacial module technology with appropriate modification of front and backside efficiencies. Thus, this study can be extended to a quantitative investigation of location-specific advantages of different solar cells. Our conclusions about the latitude-dependent choice of various farms are very general and should apply to various solar cell technologies.

For future applications of our model beyond the scope of this paper, one can consider the possibility of using bifacial PV for AgropV applications, for potential reduction in cost due to reduced height associated with vertical and tracking bifacial modules [42–44]. Our model can be extended to include various types of crop-specific tracking algorithm to control shading and perform co-optimization of the farm design. The role of bifacial PV in floating solar is unclear. Also, next generation PVK-HIT tandem solar cells could improve the efficiency further [34,45,46]. Moreover, the reliability and lifetime are important concerns, especially regarding the mechanical failure of the trackers [47–50]. The practical viability of solar tracking vs. power-tracking are not fully understood. Storage-integrated solar farming is another important aspect to consider in worldwide optimization of farm design. Finally, spectral dependence of albedo can slightly modify our numerical results, however that will not change the conclusions.

The static global maps displayed in this study can be explored interactively to find data point values for each location using DEEDS – a web-based platform for scientific research [51–53].

CCRediT authorship contribution statement

M. Tahir Patel: Conceptualization, Methodology, Software, Writing - original draft, Writing - review & editing, Data curation, Visualization.
M. Sojib Ahmed: Data curation, Software, Visualization.
Hassan Imran: Writing - original draft, Software, Methodology.
Nauman Z. Butt: Supervision.
M. Ryyan Khan: Conceptualization, Methodology,

Software, Writing - review & editing. **Muhammad A. Alam:** Conceptualization, Methodology, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apenergy.2021.116478>.

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